Mayank Sharma

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Education

Indian Institute of Technology Jodhpur — Jodhpur, India
B.Tech in Engineering Science — Majors in Computer Science & Electrical Engineering — *Aug 2023 – May 2027*

Publications & Preprints

Structural Hawkes Ambiguity in Near-Critical Market Making — Preprint · 2026
***M. Sharma**, R. K. Mourya | SSRN 6777719 · DOI: 10.5281/zenodo.20341461*
- Spectral robust-control framework showing model ambiguity in a Hawkes branching matrix is economically amplified near criticality; stress-tested on LOBSTER and Binance limit-order-book data.

Does Teacher-Matching Sentiment Quality Translate into Trading Edge? — Under review · 2026
***M. Sharma**, M. Baruah | SSRN preprint; target: ACM ICAIF 2026*
- Leakage-controlled, multi-seed distillation test of FinBERT-based sentiment trading across 33 US equities; reported edges do not survive purged cross-validation and embargoed splits.

Simplex-Ratio Open-Set Recognition Beyond the Simplex Regime — Under review · 2026
***M. Sharma** et al. | target: Transactions on Machine Learning Research (TMLR)*
- Extends regular-simplex open-set analysis to the low-dimensional regime $d < C - 1$; derives rejection geometry, threshold selection, and graceful-degradation guarantees with cross-domain validation.

Low-Rank Test-Time Adaptation for Tabular Models — Under review · 2026
***M. Sharma**, P. Mazumder | target: NeurIPS 2026 Workshop (working title)*
- LoRA-style low-rank FTTA cutting trainable adaptation parameters by 96.8%–98.1% versus full-alpha while holding near-parity accuracy; exact base-model behavior when adaptation is disabled.

ZAK-OTFS Receiver Design for Doubly-Dispersive Channels — In preparation · 2026
***M. Sharma** et al. | target: IEEE conference (ICC / GLOBECOM) (working title)*
- Equalization and receiver design for ZAK-OTFS under high-mobility doubly-dispersive channels, with regression-tested reproducible experiments.

Research Experience

Undergraduate Researcher — May 2025 – Present
University of Birmingham (Dubai Campus) | Advisor: Prof. Monita Baruah
- Built a leakage-safe financial NLP framework using teacher–student distillation (FinBERT → RATAT with temporal embeddings), with strict out-of-fold training and embargoed time-series splits across multiple equity and crypto assets.
- Constructed daily sentiment-factor pipelines and measured forecasting impact under purged cross-validation with Diebold–Mariano and Clark–West tests, isolating true incremental signal beyond price-based baselines.
- Ran asset-level robustness and degradation analysis (class behavior, regime sensitivity, cost-aware backtests), showing sentiment effectiveness is strongly market-structure dependent.

Independent Research — Quantitative Finance & Market Microstructure — 2025 – 2026
Indian Institute of Technology Jodhpur | with R. K. Mourya
- Derived a spectral robust-control framework for market making under ambiguity in Hawkes order-flow models, showing branching-matrix uncertainty is amplified through the resolvent near criticality.
- Built reproducible stress-test pipelines (common-random-number Ogata simulation) and calibrated public Hawkes models on LOBSTER and Binance limit-order-book data; released as a preprint with a full reproducibility package.

Undergraduate Researcher — Deep Learning (Test-Time Adaptation) — Dec 2025 – Present
Indian Institute of Technology Jodhpur | Advisor: Prof. Pratik Mazumder
- Implemented LoRA-style low-rank "alpha" FTTA in PyTorch with frozen pretrained weights and exact base-model behavior when adaptation is disabled.
- Reduced trainable adaptation parameters by 96.8%–98.1% versus full-alpha across MLP checkpoints while maintaining near-parity accuracy on HELOC and Diabetes.
- Built batch-level FTTA analytics (with- vs. without-alpha accuracy, per-batch delta, shift metrics) with automated CSV reporting for reproducible evaluation.

Independent Research — Open-Set Recognition Theory — 2025 – 2026
Indian Institute of Technology Jodhpur | with collaborators
- Extended regular-simplex open-set recognition to the low-dimensional regime $d < C - 1$; derived rejection geometry, threshold-selection rules, and graceful-degradation guarantees.
- Validated theory against reproduced medical-imaging baselines and cross-domain experiments (AUROC, OSCR, FAR–KRR tradeoffs, symmetry-defect metrics).

Researcher — Graph Theory & Algorithms (Feedback Vertex Set) Aug 2024 – Nov 2024

Indian Institute of Technology Jodhpur | *Advisor: Prof. Lawqueen Kanesh*

- Analyzed NP-hard graph optimization problems and derived formal proofs for time-complexity bounds on large-scale network topologies.
- Implemented Fixed-Parameter Tractable (FPT) algorithms including kernelization; benchmarked empirically against greedy baselines.

Research — Wireless Communications (ZAK-OTFS) 2026 – Present

Indian Institute of Technology Jodhpur | *Advisor: [fill in]*

- Designing receiver and equalization methods for ZAK-OTFS under high-mobility, doubly-dispersive channels, with regression-tested reproducible experiments.

Industry Experience

Quantitative Data Engineering Intern May 2025 – Jul 2025

Detachmint Ventures Pvt. Ltd. *Hybrid, India*

- Optimized signal-processing pipelines for time-series data via noise filtering (seasonality/liquidity filters), improving predictive stability on out-of-sample data.
- Designed volatility-adaptive statistical models (EWMA, GARCH) to reduce tail risk in stochastic systems, validated through extensive historical simulation.
- Engineered a high-throughput pipeline processing 80+ parallel high-frequency (1-min) data streams in Python/Pandas, tuned for memory efficiency and latency.

BRAIN Research Consultant Dec 2023 – Sep 2025

WorldQuant *Remote*

- Researched and backtested 2,000+ systematic alpha signals on the WorldQuant BRAIN platform using statistical and ML-driven time-series techniques across 10+ years of equity data (simulated Sharpe > 3 in backtest).
- Validated strategies under strict in-sample/out-of-sample separation, prioritizing out-of-sample stability and low signal decay over peak in-sample fit.

Selected Projects

Stochastic Optimal Control Simulation & PDE Solving Nov 2024

Independent Project — Python, NumPy, Matplotlib

- Developed a numerical Hamilton–Jacobi–Bellman (HJB) PDE solver to derive optimal control policies; simulated stochastic inventory dynamics and boundary controls under Brownian-motion constraints.

Short-Term Energy Load Forecasting using LSTM Networks Oct 2024

Independent ML Project — PyTorch, Time-Series Modeling

- LSTM forecaster for hourly energy demand with engineered temporal features; benchmarked against ARIMA and Gradient Boosting for a 22% RMSE reduction.

Achievements & Awards

- **STEMS Residential Camp (CMI), 2026** — selected among the top 40 students nationwide for a fully funded, research-oriented mathematics camp at Chennai Mathematical Institute.
- **ICPC 2025 Online Winter Challenge (Huawei)** — Global Rank 192; designed heuristics for dynamic topology control and traffic load balancing in large-scale optical networks.
- **Baruch Pre-MFE (Linear Algebra)** — selected among 40 worldwide, Distinction; advanced linear algebra & numerical methods.
- **JEE Advanced 2023**: Top 6.8% among 200,000+ candidates | **JEE Main 2023**: Top 2.1% among 1.1M+ candidates.
- **IIT Roorkee Trading Heist 2025** — placed 6th among 631 teams nationwide (Algorithmic Strategy).
- **Competitive Programming** — Codeforces Expert (rating 1736).

Technical Skills

- **Programming:** C++ (STL, memory management), Python (NumPy, PyTorch, Pybind11), SQL, Bash
- **Scientific Computing & Modeling:** Time-series analysis (ARIMA, LSTM), stochastic processes, numerical optimization, PDE solving, Monte Carlo simulation
- **ML & Statistics:** PyTorch, scikit-learn, XGBoost, pandas, NumPy, statsmodels, arch, TA-Lib
- **Tools:** Git, Linux, Jupyter, Backtrader, Streamlit, REST APIs, Pybind11, OpenSSL

Relevant Coursework

- Data Structures & Algorithms, Design & Analysis of Algorithms, Operating Systems, Computer Architecture, Database Systems, Signals & Systems, Probability/Statistics & Stochastic Processes, Machine Learning, Communication Systems, Embedded Systems, Deep Learning, Computer Vision, Statistical Inference
- *Upcoming:* Time-Series Analysis, Optimization, Autonomous Systems